Question 7

At 8 am; $f(x) = -x^3 + 6x^2 + 15x$

At 10 am; $f(2) = -(2)^3 + 6(2)^2 + 15(2)$

$= 46$ television sets

Question 8

(a) $f(0) = 32°\text{F}$

$f(100) = 212°\text{F}$

(b) $1.8x + 32 = 1220$

$x = 660°\text{C}$

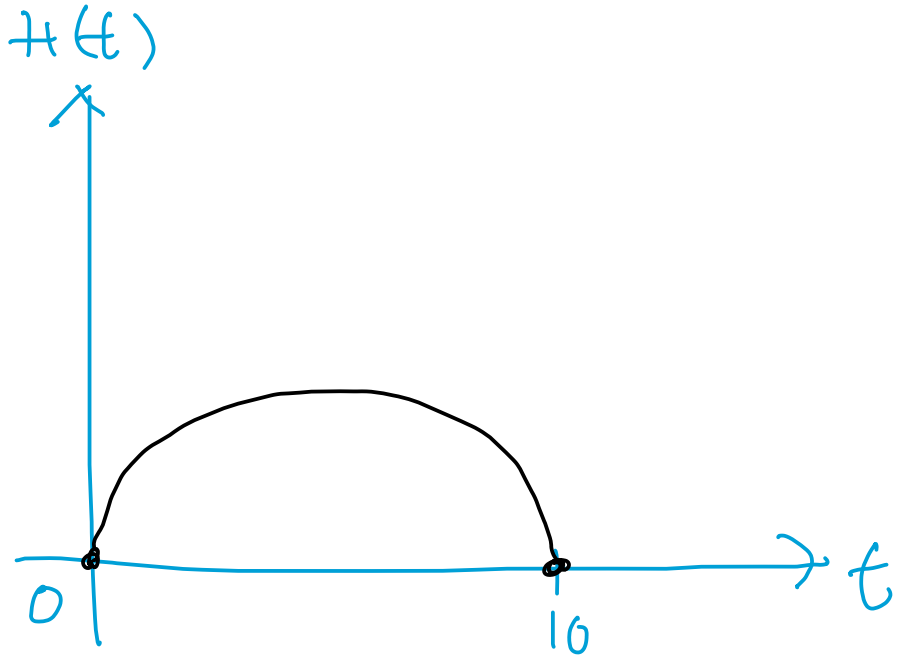
Question 9

(a) $t = 0; y = 0$

$y = 0; -16t^2 + 160t = 0$

$-16t(t - 10) = 0$

$t = 0,\ t = 10$



(b) After 10 seconds (because height is zero ft)

(c) $-16t^2 + 160t$

$= -16[(t - 5)^2 - 25]$

$= -16(t - 5)^2 + 400$

The object's maximum height is 400 ft

Question 10

(a) $Ae^0 = 10$

$A = 10$

$10e^{5k} = 8$

$e^{5k} = 0.8$

$k = \frac{1}{5}\ln(0.8)$

(b) $10e^{\left(\frac{1}{5}\ln(0.8)\right)(10)} = 6.4\ grams$

(c) $10e^{\left(\frac{1}{5}\ln(0.8)\right)(x)} = 15.53$

$x = 15.5\ years\ (to\ the\ nearest\ tenth)$